Assignment No. 1

Partial Differential Equation-MTH646

Semester: Fall 2025

Total Marks: 15

Question: 1 **Marks: 05**

The finite difference form of the one-dimensional heat equation is expressed as

$$w_{i,j+1} = w_{i-1,j} - w_{i,j} + w_{i+1,j},$$

subject to the boundary and initial conditions

$$w(0,t) = w_{0,j} = 0, \quad w(1,t) = w_{20,j} = 0,$$

and

$$w(x,0) = w_{i,0} = \sin(\pi x_i).$$

If the spatial domain $0 \leq x \leq 1$ is divided into 20 equal parts, giving $h = 0.05$, determine the values of $w_{1,1}$ and $w_{2,1}$.

Solution:

Initial Conditions (j=0)

The initial condition is given by:

$$\omega(x,0) = \omega_{i,0} = \sin(\pi x_i)$$

The spatial domain $0 \leq x \leq 1$ is divided into 20 equal parts, so the spatial step is h = 0.05 The spatial nodes are xi = i. h = 0.05i.

For

$$i = 1$$

$$\omega_{1,0} = \sin(\pi x_1) = \sin(\pi.0.05.1) = \sin(0.05\pi)$$

$$i = 2$$

$$\omega_{2,0} = \sin(\pi x_2) = \sin(\pi.0.05.2) = \sin(0.1\pi)$$

Boundary Conditions

The **boundary conditions** at x=0 and x=1are:

$$\omega(0,t) = \omega_{0,j} = 0$$

$$\omega(1,t) = \omega_{20,j} = 0$$

This is true for all time steps j≥ 0. Specifically, for the time step j=0 $\omega_{0,0} = 0$

Calculating: $\omega_{1,1}$

Use the finite difference formula with i=1 and j=0

$\underline{\omega_{1,0+1} = \omega_{1-1,0} - \omega_{1,0} + \omega_{1+1,0}}$

$\omega_{1,+1} = \omega_{0,0} - \omega_{1,0} + \omega_{2,0}$

putting Value :

$\omega_{1,1} = 0 - \sin(0.05\pi) + \sin(0.1\pi)$

$\omega_{1,1} = 0 - \sin(0.1\pi) - \sin(0.05\pi)$

Calculating $\omega_{2,1}$

Use the finite difference formula with $i = 2$ *and* $j = 0$

$\omega_{2,0+1} = \omega_{2-1,0} - \omega_{2,0} + \omega_{2+1,0}$

$\omega_{2,+1} = \omega_{1,0} - \omega_{2,0} + \omega_{3,0} \rightarrow (1)$

$\omega_{3,0}$?

$\omega_{3,0} = \sin(\pi x_3) = \sin(\pi.0.05.3) = \sin(0.15\pi)$

Question: 2	**Marks: 05**

The finite-difference form of the one-dimensional heat equation is

$$w_{i,j+1} = w_{i-1,j} - w_{i,j} + w_{i+1,j},$$

subject to the boundary and initial conditions

$$w(0,t) = w_{0,j} = 0, \quad w(1,t) = w_{20,j} = 0,$$

and

$$w(x,0) = w_{i,0} = \sin(\pi x_i).$$

If the domain $0 \le x \le 1$ is divided into 20 equal parts (so $h = 0.05$), find the values of $w_{5,1}$ and $w_{6,1}$.

Solution:

$$w_{i,j+1} = w_{i-1,j} - w_{i,j} + w_{i+1,j}$$

Calculating $w_{5,1}$

Use the finite difference formula with $i = 5$ *and* $j = 0$

$$w_{5,0+1} = w_{5-1,0} - w_{5,0} + w_{5+1,0}$$

$$w_{5,+1} = w_{4,0} - w_{5,0} + w_{6,0}$$

substitute the initial condition $w_{i=0} = \sin(0.05\pi i)$

$$w_{4,0} = \sin(0.05\pi.4) = \sin(0.20\pi)$$

$$w_{5,0} = \sin(0.05\pi.5) = \sin(0.25\pi)$$

$$w_{6,0} = \sin(0.05\pi.6) = \sin(0.30\pi)$$

$$w_{5,1} = \sin(0.20\pi) - \sin(0.25\pi) + \sin(0.30\pi)$$

Calculating $w_{6,1}$

Use the finite difference formula with $i = 6$ *and* $i = 0$

Question: 3	**Marks: 05**

The analytical solution of the one-dimensional heat equation is expressed as

$$w(x,t) = e^{-a^2\pi^2 t}\sin(\pi x).$$

Determine the values of $w(x_7, t_1)$, $w(x_8, t_1)$, and $w(x_9, t_1)$ when the parameters are $a^2 = 1$, $h = 0.05$, and $k = 0.01$.

Here, $x_j = jh$ and $t_n = nk$ represent the spatial and temporal grid points, respectively.

Solution:

1)

Formula:

$$w(x,t) = e^{-a^2\pi^2 t}\sin(\pi x)$$

$a^2 = 1$

$$h = 0.05, k = 0.01$$

now Grid point value:

$$x_j = jh, t_n = nk$$

time: we need value at $t_1 (n = 1)$

$$t_1 = 1, k = 0.01$$

Spatial points:

$$x_7 = 7.h = 7.0.05 = 0.35$$

$$x_8 = 8.h = 8.0.05 = 0.40$$

$$x_9 = 9.h = 9.0.05 = 0.45$$


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Booking Start
I Can Manage All
Activities Like
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2)

General Analytical Expression

Substitude :

$a^2=1$

$t = t_1 = 0.01$ into *the solution formula*

$w(x,t_1) = e^{-(1)\pi^2(0.01)}\sin(\pi x) = e^{-\pi^2(0.01)}\sin(\pi x)$

since $e^{-\pi^2(0.01)}$ *is for* t_1

$e^{-\pi^2(0.01)} \approx e^{-0.01.9.8696} \approx e^{-0.098696} \approx 0.9060$

$w(x,t_1) \approx 0.9060\sin(\pi x).$

3)

Calculating $w(x_j, t_1)$

substitude x_7, x_8, x_9

$w(x_7,t_1) = e^{-(0.01)\pi^2}\sin(0.35\pi) \approx 0.9060.(0.8910) \approx 0.8073$

$w(x_8,t_1) = e^{-(0.01)\pi^2}\sin(0.40\pi) \approx 0.9060.(0.9511) \approx 0.8617$

$w(x_9,t_1) = e^{-(0.01)\pi^2}\sin(0.45\pi) \approx 0.9060.(0.9877) \approx 0.8949$